Haoyi Duan

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Education Background

Stanford University, California, USA **09/23-present**
M.S. Electrical Engineering

Zhejiang University, Zhejiang, China **09/2019-06/2023**
B.Eng. Computer Science and Technology, **Chu Kochen Honors College**
GPA: **3.97**/4.00 (**91.38**/100) The 3rd year GPA **3.99**/4.00 (**93.17**/100)
Ranking: **1st** / 227

- ✧ **National Scholarship** awarded in 2021-2022 (TOP 1%)
- ✧ **First-Class Scholarship** awarded in 2019-2020, 2020-2021, and 2021-2022 (TOP 2%)

National University of Singapore, Singapore **05/2022-07/2022**
School of Computing Summer Workshop </>

- ✧ Took the course **Real-Time Graphics Rendering** (A+)
- ✧ Programmed a ray-tracing system and won the Third Prize (Ranking: 3/18)

Publications & Patents

Cross-modal Prompts: Adapting Large Pre-trained Models for Audio-Visual Downstream Tasks
Haoyi Duan, Yan Xia, Mingze Zhou, Li Tang, Jieming Zhu, Zhou Zhao, *NeurIPS'2023 (Under Review)*

Time as Prompt for Geography-aware Next Location Recommendation </>
Yan Luo, **Haoyi Duan**, Ye Liu, Chung Fu-Lai, *CIKM'2023*

Beyond Two-Tower Matching: Learning Sparse Retrievable Interaction Models for Recommendation </>
Liangcai Su, Fan Yan, Jieming Zhu, Xi Xiao, **Haoyi Duan**, Zhou Zhao, Zhenhua Dong, Ruiming Tang, *SIGIR'2023*

LaPE: Layer-adaptive Position Embedding for Vision Transformers with Independent Layer Normalization
Runyi Yu, Zhennan Wang, Yinhuai Wang, Kehan Li, Chang Liu, **Haoyi Duan**, Xiangyang Ji, Jie Chen, *ICCV'2023*

MPI, OpenMP, and Taichi Parallel Programming Languages </>
Haoyi Duan, *Electronic Components and Information Technology 2022*

Lock-in Amplifier Measures Weak Sound Pressure Signals </>
Haoyi Duan, *Computer Knowledge and Technology 2021*

OpenGL-based "OpenCity" Game, national patent no.:2022SR0805651, Copyright Protection Center of China

Internship

Intern Algorithm Engineer, **Huawei Noah's Ark Lab, China** **06/2022-08/2023**

- ✧ Work with Search & Recommendation Group
- ✧ Contribute to high recommendation and search accuracy with machine learning and NLP technologies in cross-modal domain

Research Projects

Adapting Large Pre-trained Models for Audio-Visual Downstream Tasks **12/2022-08/2023**
Supervisor: Prof. Zhou Zhao, Zhejiang University DCD Lab

- ✧ Proposed a novel Dual-Guided Spatial-Channel-Temporal attention mechanism, which leverages audio and visual modalities as soft prompts to dynamically adjust the parameters of pre-trained models based on the current multi-modal input features
- ✧ Achieved state-of-the-art results across multiple downstream tasks; exhibited promising performance in challenging few-shot and zero-shot scenarios

Time as Prompt for Next Location Recommendation </> **08/2022-11/2022**
Supervisor: Prof. Chung Fu-Lai, The Hong Kong Polytechnic University

- ✧ Proposed a Temporal Prompt-based and Geography-aware (TPG) framework
- ✧ Designed Temporal prompt to incorporate temporal information of next location and then devised Shifted window to augment geographic data for addressing the hard boundary problem
- ✧ Improved the effectiveness and superiority of recommendation under various settings, with a creative function of interval prediction, i.e., predicting the location that a user wants to go to at a certain time while the most recent check-in behavioral data is masked

Pre-training Audio Representations **03/2022-08/2022**

Supervisor: Prof. Zhou Zhao, Zhejiang University DCD Lab

- ✧ Improved the effectiveness of audio-related downstream tasks from massive amounts of unlabeled data through pre-training and high-quality audio representations
- ✧ Achieved state-of-the-art level in both cover recognition and sound scene classification tasks with a novel self-supervised audio representation learning framework, Audio-PAW, that contains three key components--Hierarchical Audio Cropping, Interaction Unit and Hierarchical contrastive
- ✧ Brought 14% accuracy improvements on music tagging; launched in Huawei's music recommendation services

VQA Training Based on MindSpore </> **02/2022-06/2022**

Supervisor: Prof. Siliang Tang, Zhejiang University DCD Lab

- ✧ Output the correct answer from the inputted question that contained an image and text in the form of natural language
- ✧ Applied a bi-channel image plus text model to obtain the probability of each answer to the question
- ✧ Used the Stack Attention model to filter the noise and obtain the highly answer-related region
- ✧ Realized a joint embedding of inputting an image and a question and a multi-label classifier based on a fixed set of candidate answers with the Top-down Attention model

Attribute Classification of COVID-19-Related Tweets, Student Research Training Program </> **04/2021-05/2022**

Supervisor: Prof. Siliang Tang, Zhejiang University DCD Lab

- ✧ Created a neural network for the automated identification of a COVID-19 tweet's binary attribute
- ✧ Data Augmentation by mutual translation on different language training datasets in the data preprocessing stage
- ✧ Used Bert, RoBERTa, and XLM-RoBERTa to build pre-training models and utilized LSTM+Attention, MultiHead Attention, the TextCNN model, and the adjustable Loss function in creating the classification models
- ✧ Trained 12 models that far surpass the best average F1-score of 89.7%, optimized the training result by adopting voting mechanism and reached an ultimate best-vote of 93.54%

Course Projects

"OpenCity" Game Project (Supervisor: Prof. Hongzhi Wu) </> **09/2021-01/2022**

- ✧ Developed the "OpenCity" inspired by "SimCity"
- ✧ Rendered images within the OpenGL framework

Sokoban Game Software Programmed in x86 (Supervisor: Dr. Honghuan Bai) </> **05/2021-07/2021**

- ✧ Converted C language of a puzzle video game genre to assembly language versions

"Baba Is You" Game (Supervisor: Dr. Haifeng Liu, Dr. Qingsong Shi, and Prof. Wenzhi Chen) </> **09/2020-01/2021**

- ✧ Programmed the replica game "Baba Is You" on FPGA based on Verilog
- ✧ Designed the single-cycle, multi-cycle, pipelining CPU with the functions such as Cache buffer and interrupt trigger

Society Memberships & Extracurricular Activities

Head of Saxophone Voice, Zhejiang University Marching Brass Band **09/2019-06/2023**

- ✧ Organize and join the regular training and performances (as a player of Level 10 Certificate in Saxophone), including the National Day Flag Raising, the Student Festival Tour, Zhejiang University's 2019 New Year's Gala

Player, Sports Games of Zhejiang University **10/2020-05/2022**

- ✧ The 1st Place in Sanhao Cup Zhejiang University Rumba Dance Competition 05/2022
- ✧ The Final Four in the Basketball Tournament of Chu Kochen Honors College 05/2021
- ✧ The 8th Place in 400m hurdles in Zhejiang University Sports Meeting 10/2020

Officer of Cultural and Creative Center, CKC Housing Committee **09/2020-09/2021**

- ✧ Planned cultural festivals and attractive salons

Volunteer, Zhejiang University Volunteer Association **09/2019-06/2021**

- ✧ Participated in over 120 hours of volunteer activities such as study room management, information session at High-School Alma Mater, organizing books at the children's branch of Hangzhou Library, and taking the referee role in the university sports meeting

Senior Tutors Program of Chu Kochen Honors College, Zhejiang University **08/2020-09/2020**

- ✧ Helped first-year students adapt to university life and study and won the Excellent Tutor Award in May 2021

Languages & Skills

Languages: Chinese (native) English (proficient) **TOEFL: 108(S23) GRE: (V155+Q169+3.5)**

Programming: C/C++, Python, CUDA, mini SQL, Java, Verilog, x86, Shell

Tools/Software: Pytorch, OpenGL, MindSpore, MySQL, LATEX, Vivado, Adobe PS/Pr

Prizes & Honors

Excellent Developer, awarded by Huawei on outstanding contribution to the open-source community 2022

Bronze Prize in the 8th "Intel Cup" Parallel Application Challenge 2020

Third Prize of the College Students Physics Innovation (Theoretical) Competition 2020